

# **Scenario-Based Assessment**

## **AI Healthcare Assistant using Hugging Face & Prompt Engineering**

### **Background**

You have recently joined MediAssist AI Pvt. Ltd. as a Generative AI Engineer.

The company is developing an AI assistant for hospitals that helps doctors summarize patient reports, answer clinical questions, and generate discharge summaries.

The current chatbot has several problems:

- Gives inconsistent responses
- Hallucinates medical information
- Produces unstructured outputs
- Cannot switch between different LLMs
- Difficult to maintain because prompts are poorly written

Your manager asks you to redesign the solution using Hugging Face Transformers and Prompt Engineering best practices.

### **Your Tasks**

#### **Task 1 — Hugging Face Integration**

Build a chatbot using Hugging Face.

Requirements:

- Load any instruction-tuned model from Hugging Face
- Use AutoTokenizer
- Use AutoModelForCausalLM
- Generate responses using the Transformers pipeline
- Allow the user to ask medical questions interactively

Example:

User:

Patient has fever for 3 days.

Assistant:

Possible causes...

## **Task 2 — Prompt Engineering**

The following prompt is performing poorly.

Tell me about diabetes.

Rewrite it using:

- Role Prompting
- Context
- Constraints
- Output Formatting

The response should contain:

- Disease Overview
- Symptoms
- Risk Factors
- Prevention
- When to Consult a Doctor

Output must be in Markdown.

## **Task 3 — Few-Shot Prompting**

Create a Few-Shot prompt that converts clinical notes into structured summaries.

Example

Input

Patient has cough and fever.

Output

Symptoms:

- Cough
- Fever

Severity:

Moderate

Provide at least three examples before the final input.

## **Task 4 — Chain of Thought Prompting**

A patient has

- Fever
- Cough
- Oxygen Level = 88%

Design a prompt that encourages the LLM to reason step-by-step before providing the final recommendation.

*(Prompt design only; do not reveal the model's internal reasoning.)*

## **Task 5 — Structured Output**

Design a prompt that forces the LLM to return JSON in the following format:

```
{  
  "disease": "",  
  "symptoms": [],  
  "risk_level": "",  
  "recommendation": ""
```

}

## **Task 6 — Model Comparison**

Compare any two Hugging Face models.

Include:

- Model Name
- Parameters
- Best Use Case
- Advantages
- Limitations

Present the comparison in a table.

## **Task 7 — Prompt Optimization**

The AI produces the following response:

Maybe the patient has malaria.

Or maybe dengue.

Or maybe COVID.

I am not sure.

Explain:

- Why is the response poor?
- Which Prompt Engineering techniques would improve it?
- Rewrite the prompt to reduce hallucinations.

## **Bonus Challenge**

Your manager wants the chatbot to support multiple departments.

The assistant should automatically behave differently for:

- Doctor
- Nurse
- Medical Student
- Patient

Create a prompt template that changes the response based on the user's role while maintaining the same underlying Hugging Face model.

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## **Deliverables**

Students must submit:

1. Python notebook implementing the Hugging Face chatbot.
2. Well-designed prompts for each task.
3. Screenshots of chatbot interactions.
4. A short report (2–3 pages) explaining:
  - Model selection
  - Prompt engineering techniques used
  - Challenges faced
  - Improvements made after prompt refinement